A Ternary Word Attaining the Abelian Maximal Pattern Complexity Bound at Pattern Size Three

Carptopus

carptopus@163.com

25 August 2026

v0.1-beta; not peer reviewed

Abstract

Kamae, Widmer and Zamboni proved that every recurrent word over an alphabet of size r which is aperiodic by projection has Abelian maximal pattern complexity at least $(r-1)k+1$ at pattern size k. Their introduction left open whether this bound is attainable when $r \geq 3$ and $k \geq 3$. We settle the smallest unresolved parameter pair under the literature search described below. Let α be the fixed point beginning in 0 of the constant-length substitution

$$0 \mapsto 001, \qquad 1 \mapsto 020, \qquad 2 \mapsto 000.$$

We prove that α is recurrent and aperiodic by projection and that

$$p^{*\mathrm{ab}}_{\alpha}(3) = 7.$$

The upper bound is computer-assisted but exact. Sparse three-point languages obey a residue recursion modulo 3. A 3×3 patch of neighboring languages makes this recursion finite-state; the closure contains 938 states and covers every pair of nonnegative offsets through their base-3 digits. Every central state has at most seven Parikh vectors, while the pattern $\{0, 2, 9\}$ realizes seven. The verification program is supplied with a canonical transition-table hash and fail-closed checks.

Keywords. Abelian maximal pattern complexity; infinite word; primitive substitution; aperiodic by projection; automatic sequence; computer-assisted proof.

2020 Mathematics Subject Classification. 68R15, 37B10.

1 Introduction

Let $\mathbb{A}$ be a finite alphabet and let $\alpha = \alpha_0\alpha_1\alpha_2 \cdots \in \mathbb{A}^{\mathbb{N}}$. For a finite set $S = \{s_1 < \cdots < s_k\} \subset \mathbb{N}$, an S-factor is

$$\alpha[n + S] = \alpha_{n+s_1}\alpha_{n+s_2} \cdots \alpha_{n+s_k}.$$

Two finite words are Abelian equivalent when they have the same Parikh vector. Write $p^{\mathrm{ab}}_{\alpha}(S)$ for the number of Abelian classes among all S-factors and

$$p^{*\mathrm{ab}}_{\alpha}(k) = \sup_{|S|=k} p^{\mathrm{ab}}_{\alpha}(S).$$

We call α recurrent if each of its finite factors occurs infinitely often. For a nonempty proper subset $B \subsetneq \mathbb{A}$, let $\mathbf{1}_B(\alpha)$ be the binary word whose n-th term records whether $\alpha_n \in B$. The word α is *aperiodic by projection* if $\mathbf{1}_B(\alpha)$ is not eventually periodic for every such B.

Kamae, Widmer and Zamboni [1, Theorem 4] proved that if α is recurrent and aperiodic by projection over an r-letter alphabet, then

$$p^{*\mathrm{ab}}_{\alpha}(k) \geq (r-1)k + 1. \tag{1}$$

For binary words equality is automatic, and for arbitrary r they gave equality examples at $k = 2$. Their introduction records the pointwise sharpness problem for $r \geq 3, k \geq 3$. Their Remark 4.2 asks the stronger and still separate question whether one word can attain equality for every k. The survey of Fici and Puzynina [2] lists only the previously known equality cases. The present note settles $r = 3, k = 3$, the smallest unresolved parameter pair under our current search; it does not answer the stronger all-k question.

Our construction is a primitive constant-length substitution rather than the 2-adic Toeplitz family used in [1] for $k = 2$. The main difficulty is not finding seven classes for one pattern, but proving that no three-point pattern has eight. A bounded scan over offsets cannot establish that assertion. We instead give an exact digit automaton whose finite closure covers all offset pairs.

2 Main result

Define the length-three substitution σ on $\mathbb{A} = \{0, 1, 2\}$ by

$$\sigma(0) = 001, \qquad \sigma(1) = 020, \qquad \sigma(2) = 000. \tag{2}$$

Since $\sigma(0)$ begins with 0, the limit

$$\alpha = \lim_{m\to\infty} \sigma^m(0)$$

is a well-defined one-sided fixed point.

Theorem 2.1. *The word α is recurrent and aperiodic by projection, and*

$$\boxed{p^{*\mathrm{ab}}_{\alpha}(3) = 7.}$$

Thus the lower bound (1) *is sharp for $r = 3, k = 3$.*

The proof occupies Sections 3–5. Section 3 verifies the dynamical hypotheses, Section 4 gives the exact sparse-language recursion, and Section 5 converts that recursion into a finite certificate covering every pattern.

3 Recurrence and aperiodicity by projection

Proposition 3.1. *The substitution σ is primitive. Consequently its fixed point α is uniformly recurrent.*

Proof. The word $\sigma(0)$ contains $0, 1$, $\sigma(1)$ contains $0, 2$, and $\sigma(2)$ contains 0. Iterating at most three further times therefore makes every letter occur in the image of every letter. Hence the incidence matrix of σ is primitive. The standard primitive-substitution theorem [3] implies uniform recurrence of the fixed point. □

For $i \in \{0, 1, 2\}$, put

$$\chi_i(n) = \mathbf{1}_{\{\alpha_n = i\}}.$$

Reading (2) coordinatewise gives

$$\begin{aligned}
\alpha_{3n} &= 0, \\
\chi_2(3n+1) &= \chi_1(n), \\
\chi_1(3n+2) &= \chi_0(n), \\
\chi_0(3n+2) &= 1 - \chi_0(n).
\end{aligned} \tag{3}$$

Moreover, χ_2 is supported on $1 \pmod 3$, while χ_1 is supported on $2 \pmod 3$.

Proposition 3.2. *The word α is aperiodic by projection.*

Proof. Primitivity implies that each letter occurs infinitely often. Suppose first that χ_0 is eventually periodic with positive period p. If $3 \nmid p$, the identity $\chi_0(3n) = 1$, together with $\gcd(3, p) = 1$, forces χ_0 eventually to be identically one. This contradicts the infinite occurrence of 1 and 2. If $p = 3q$, then the last identity in (3) shows that q is also an eventual period of χ_0. Removing all powers of 3 from p reduces to the preceding contradiction.

If χ_1 is eventually periodic, its support in one residue class modulo 3, together with infinitely many ones, forces every eventual period to be divisible by 3. The third identity in (3) then transfers an eventual period divided by 3 to χ_0, which is impossible. The same argument using the second identity transfers eventual periodicity of χ_2 first to χ_1 and then to χ_0.

Every nonempty proper subset of a three-letter alphabet is either a singleton or the complement of a singleton. Its indicator is therefore one of the χ_i, or its complement. No nontrivial projection is eventually periodic. □

4 Exact recursion for sparse languages

For an ordered integer tuple $C = (c_1, \ldots, c_m)$, define

$$\mathcal{L}(C) = \{(\alpha_{n+c_1}, \ldots, \alpha_{n+c_m}) : \text{the tuple occurs for arbitrarily large } n \text{ with } n + c_i \geq 0 \text{ for all } i\}.$$

Adding the same integer to every coordinate does not change $\mathcal{L}(C)$. Uniform recurrence also ensures that deleting a finite initial segment does not change this language. Let

$$f_d(x) = \sigma(x)_d, \qquad d \in \{0, 1, 2\}.$$

Lemma 4.1 (residue recursion)**.** *Normalize C by subtracting $\min_i c_i$. For each $t \in \{0, 1, 2\}$, put*

$$q_i = \left\lfloor \frac{t + c_i}{3} \right\rfloor, \qquad d_i \equiv t + c_i \pmod 3,$$

and normalize $Q_t = (q_i)_i$ by a common translation. Then

$$\mathcal{L}(C) = \bigcup_{t=0}^{2} \{(f_{d_1}(x_1), \ldots, f_{d_m}(x_m)) : (x_1, \ldots, x_m) \in \mathcal{L}(Q_t)\}. \tag{4}$$

Proof. Write the translation origin as $n = 3u + t$. Each sampled coordinate then lies in position d_i of the substituted letter α_{u+q_i}. Conversely, every occurrence of a child tuple at an arbitrarily large u produces an occurrence in the corresponding residue class. Common normalization only shifts u, and recurrence removes the finite boundary. No coordinate reordering is performed. □

The recursion terminates at span one. The exact two-letter factor set is

$$\mathcal{F}_\alpha(2) = \{00, 01, 02, 10, 20\}. \tag{5}$$

Indeed, internal factors of the three substitution images give $00, 01, 02, 20$; every image starts in 0, while its final letter is 0 or 1, so crossing a substitution boundary adds only 00 and 10.

5 The finite patch certificate

For integers a, b, define the nine-cell patch

$$K(a,b) = \big(\mathcal{L}(0, a+i, b+j)\big)_{i,j\in\{-1,0,1\}}. \tag{6}$$

Lemma 5.1 (digit transition)**.** *For every $x, y \in \{0,1,2\}$, there is an explicit deterministic map $\Phi_{x,y}$ such that*

$$K(3a+x, 3b+y) = \Phi_{x,y}(K(a,b)). \tag{7}$$

Write $K_{u,v}$ for the (u,v)-cell of a patch. For each target cell $(i,j) \in \{-1,0,1\}^2$ and $t \in \{0,1,2\}$, set

$$u_t = \left\lfloor \frac{t+x+i}{3} \right\rfloor, \qquad v_t = \left\lfloor \frac{t+y+j}{3} \right\rfloor,$$

and

$$d_t = (t,\ (t+x+i) \bmod 3,\ (t+y+j) \bmod 3).$$

Then

$$[\Phi_{x,y}(K)]_{i,j} = \bigcup_{t=0}^{2} \{(f_{d_{t,0}}(z_0), f_{d_{t,1}}(z_1), f_{d_{t,2}}(z_2)) : (z_0, z_1, z_2) \in K_{u_t,v_t}\}. \tag{8}$$

Proof. Apply (4) to each of the nine target cells. After division by 3, each quotient offset differs from a or b by an element of $\{-1,0,1\}$. Thus every required child language is already a cell of $K(a,b)$; the output digits determine the coordinate maps f_d. □

Starting from $K(0,0)$, close under the nine maps $\Phi_{x,y}$. The fail-closed verification obtains the following certificate.

Theorem 5.2 (finite closure certificate)**.** *The transition closure contains 938 patch states and 8442 directed labelled transitions. For the central language of a state, the distribution of the number of Parikh vectors is*

number of vectors	*number of states*
3	*3*
4	*246*
5	*223*
6	*285*
7	*181*

The maximum is seven. The canonical JSON serialization of the states and all labelled transitions has SHA-256 digest

BBF0CBF7D6E5E23CCEECC71A59644AD4F5DA0D48A1A2A0A219BB292A127CDAAA.

Why this covers every pattern. Let $0 \le a, b$. Read the paired base-3 digits of a, b from most significant to least significant, padding with leading zeros. Repeatedly apply the corresponding transition $\Phi_{x,y}$ to $K(0,0)$. Equation (7) shows inductively that the final state is $K(a,b)$. Therefore this finite closure covers every nonnegative offset pair; it is not a bounded search over offsets. In particular, every three-point pattern $\{s_1 < s_2 < s_3\}$, after translation to $\{0, s_2 - s_1, s_3 - s_1\}$, has at most seven Abelian classes.

The supplied program cross-checks the Parikh images of the recursive languages against a length 3^{10} fixed-point prefix for a small offset box, checks digit transitions against direct recursive patches for $0 \leq a, b < 100$, and aborts unless all state counts, the distribution and the canonical digest match the frozen values. $\square$

Proposition 5.3 (sharp pattern)**.** *For $S = \{0, 2, 9\}$, the exact Parikh set is*

$$\{(0,2,1),(1,0,2),(1,1,1),(1,2,0),(2,0,1),(2,1,0),(3,0,0)\}.$$

Hence $p^{\mathrm{ab}}_{\alpha}(S) = 7$.

Combining Theorem 5.2 and Proposition 5.3 proves the complexity equality in Theorem 2.1. Propositions 3.1 and 3.2 verify the hypotheses of (1), providing an independent lower-bound route as well.

6 Scope and prior-work boundary

The result establishes sharpness only at $r = 3, k = 3$. It does not prove that the KWZ bound is sharp for all $r \geq 3, k \geq 3$, and it does not produce a single word attaining the bound for every k.

The candidate new contribution is the explicit ternary substitution and the all-pattern proof that its three-point Abelian maximal pattern complexity is seven. Primitive-substitution recurrence and the KWZ lower bound are standard inputs. The finite-state method is used here to certify the global upper bound; the program is part of the proof package rather than evidence from a finite sample.

The literature search covered the original paper [1], the survey [2], visible citation metadata, and targeted searches for the substitution and the exact parameter pair. No direct prior construction was found. This is not an absolute-priority claim: unindexed work, different notation and concurrent research remain possible until broader expert review.

7 Reproducibility and responsibility

The archived verification file is `verification/verify_ternary_ampc_k3.py`, with SHA-256

`7F827D487681F3F6139FBB239E89EAD0A5303C6CE853866B6F6E6E082CE78CC8`.

It uses Python 3.11 or later and only the standard library. From the entry directory, run

```
python -X utf8 verification/verify_ternary_ampc_k3.py.
```

The program regenerates the finite closure, checks the frozen counts and closure hash, and prints the sharpness witness. No random seed or external solver is used. The verification README records the expected output, file manifest and evidence boundary.

Carptopus is the responsible author and takes responsibility for the mathematical claims and the released materials. OpenAI Codex was used extensively for AI-assisted research, proof development, verification and writing. The result remains subject to public mathematical review.

References

[1] T. Kamae, S. Widmer and L. Q. Zamboni, Abelian maximal pattern complexity of words, *Ergodic Theory Dynam. Systems* 35 (2015), 142–151. doi:10.1017/etds.2013.51.

[2] G. Fici and S. Puzynina, Abelian combinatorics on words: a survey, *Comput. Sci. Rev.* 47 (2023), 100532. doi:10.1016/j.cosrev.2022.100532.

[3] M. Queffelec, *Substitution Dynamical Systems—Spectral Analysis*, second ed., Lecture Notes in Mathematics 1294, Springer, 2010.